Make sure you are computing in *radians*. We will rarely use degrees from now on.

Graphs of $\sin(x)$ and $\cos(x)$

Since $\sin(x)$ and $\cos(x)$ are defined by using our unit circle, they repeat themselves every _____. In other words:

- $\sin(x) = \sin(x + ___) = \sin(x + ___) = \ldots$ and
- $\cos(x) = \cos(x + ___) = \cos(x + ___) = \ldots$

Question Use Wolfram Alpha to draw the graphs of $\sin(x)$ and $\cos(x)$ below (on the same set of axes), with domain $[-2\pi, 2\pi]$ (You can literally type '`graph of sin(x) from -2pi to 2pi`' into Wolfram Alpha):

The Tangent Function

We defined $\tan\theta = \dfrac{\text{opp}}{\text{adj}} = \frac{\sin\theta}{\cos\theta}$. Using Wolfram Alpha, draw the graph of $\tan x$ with domain $[-2\pi, 2\pi]$ below:

Questions

1. What is the period of $\tan x$?

2. For what values of x is $\tan x$ positive? Negative? Zero?

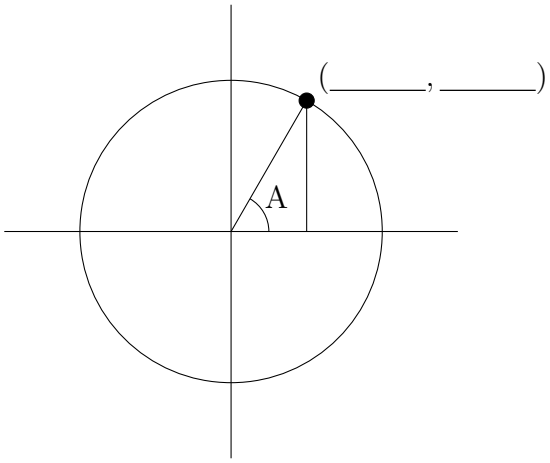
3. Where does $\tan x$ have vertical asymptotes?

4. Does $\lim_{x \to \frac{\pi}{2}} \tan x$ exist? Why or why not?

Trig Identities

Questions

1. On the unit circle below, label the coordinate of the point marked on the circle in terms of A:



2. Use Pythagoras to derive an identity relating the lengths of the sides of the right-angled triangle in the diagram to the length of its hypotenuse:

The identity you just wrote down above is the most basic trigonometric identity, from which many others can be derived. Before we do so, we'll introduce three additional trig functions:

- $\sec\theta = \frac{1}{\cos\theta} = \frac{\text{hyp}}{\text{adj}}$ - The *secant* function;
- $\csc\theta = \frac{1}{\sin\theta} = \frac{\text{hyp}}{\text{opp}}$ - The *cosecant* function;
- $\cot\theta = \frac{1}{\tan\theta} = \frac{\text{adj}}{\text{opp}} = \frac{\cos\theta}{\sin\theta}$ - The *cotangent* function.

Question Using the identity you derived above, show that the following identities are true:

- $1 + \cot^2 x = \csc^2 x$
- $\tan^2 x + 1 = \sec^2 x$

Angle Addition Formulas

We will not prove the following, though they are true and will be very useful:

- $\sin(A + B) = \sin A \cos B + \cos A \sin B$
- $\sin(A - B) = \sin A \cos B - \cos A \sin B$
- $\cos(A + B) = \cos A \cos B - \sin A \sin B$
- $\cos(A - B) = \cos A \cos B + \sin A \sin B$

Questions

1. Use the formulas above to show that $\sin(2A) = 2\sin A \cos A$.

2. Show that $\cos(\frac{\pi}{2} - x) = \sin(x)$.

You will see more of these on the homework...

An Important Limit

Using Wolfram Alpha, draw the graph of $f(x) = \frac{\sin x}{x}$ below:

Questions

1. Is $f(x)$ defined at $x = 0$? Why or why not?

2. By plugging in numbers near 0, complete the following:

$$\lim_{x \to 0} \frac{\sin x}{x} = _____.$$

Question Using Wolfram Alpha, draw the graph of $g(x) = \frac{\cos x - 1}{x}$ below:

Questions

1. Is $g(x)$ defined at $x = 0$? Why or why not?

2. By plugging in numbers near 0, complete the following:

$$\lim_{x \to 0} \frac{\cos x - 1}{x} = _____.$$

Again, we will not prove that the limit you wrote down above is true, but we will use them when doing calculus with trig functions.